

Week 2 Internship Progress Report

City Scale Retrofitting Analysis for Ahmedabad

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1 Introduction

During Week 2 of the internship at SWA Consultancy, the primary focus was on understanding the practical workflow involved in Urban Building Energy Modeling (UBEM), GIS-based dataset preparation, satellite imagery analysis, and the workflow of CityBES for city-scale building energy analysis.

The learning activities mainly included:

- Exploration of the CityBES platform
- Understanding UBEM workflows
- Learning GIS-based dataset preparation
- Exploring satellite imagery sources
- Initial exploration of QGIS
- Understanding GeoJSON datasets and building archetypes

2 Exploration of CityBES

CityBES (City Building Energy Saver) is an Urban Building Energy Modeling platform used to analyze building energy performance and retrofit scenarios at city scale.

The following concepts were studied during exploration of the CityBES platform:

- Urban Building Energy Modeling (UBEM)
- Building energy simulations
- Baseline energy simulations
- Retrofit analysis workflows
- Building archetypes
- Energy performance comparison methods

2.1 Platform Features Explored

The following CityBES modules and options were explored:

- District Buildings
- Modeling and Analysis
- Urban Environment
- District Energy
- Building Filters
- Color Buildings By
- Result Options

The exploration helped in understanding how buildings are digitally represented using:

- Building geometry
- Building type
- Building height
- Floor area
- Building age

2.2 Key Understanding

CityBES converts city-scale GIS building datasets into digital building energy models for analyzing:

- Energy consumption
- Cooling loads
- Retrofit scenarios
- Building performance

The platform also helps visualize energy-related information at district and city scales.

3 Satellite Imagery and Urban Area Analysis

Satellite imagery sources were explored to understand urban morphology and spatial characteristics of Ahmedabad city.

3.1 Satellite Data Sources Explored

The following platforms were explored:

- Google Earth Pro
- VEDAS
- Bhuvan NRSC
- Copernicus Browser
- Google Earth Engine

Among these, Google Earth Pro was mainly used for urban analysis and screenshot collection.

3.2 Selected Ahmedabad Areas

The following Ahmedabad regions were selected for urban analysis:

- Navrangpura
- Maninagar
- SG Highway
- Bopal
- Naroda
- Chandkheda
- Vastrapur

3.3 Observations Made

The following urban characteristics were observed through satellite imagery analysis:

- Urban density and building compactness
- Road network patterns
- Green cover distribution
- Residential and commercial patterns
- Urban morphology variations

This analysis helped in understanding how different urban forms influence building energy behavior and cooling demand.

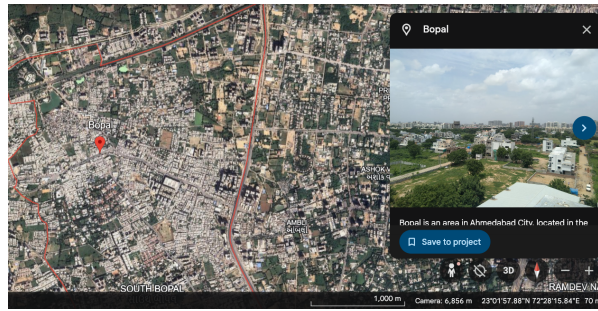


Figure 1: Bopal

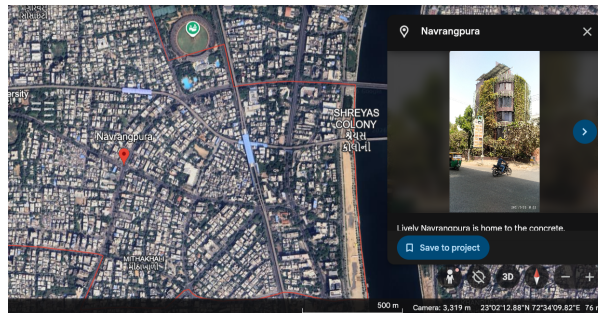


Figure 2: Navrangpura

3.4 Location Identification

Google Earth Pro was also used to identify:

- Latitude
- Longitude
- Elevation
- Satellite imagery screenshots

These spatial details are important for GIS-based dataset preparation and urban energy analysis.

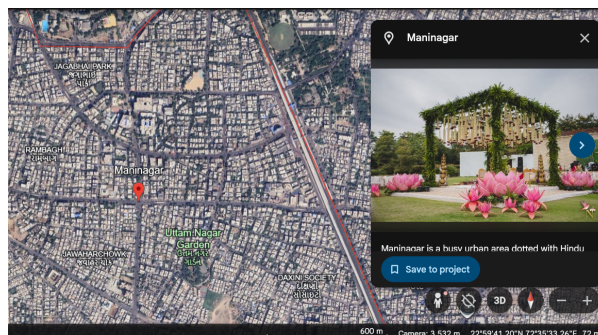


Figure 3: Maninagar

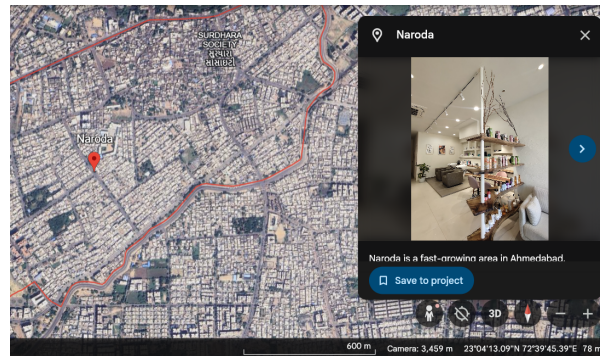


Figure 4: Naroda

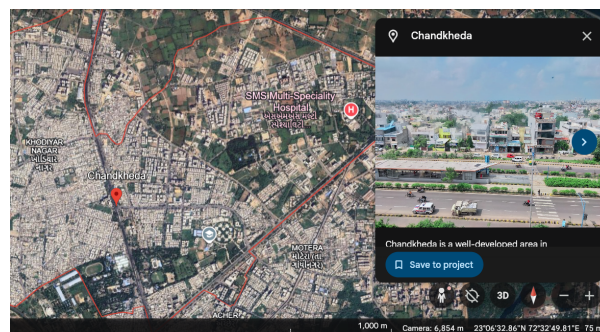


Figure 5: Chandkheda

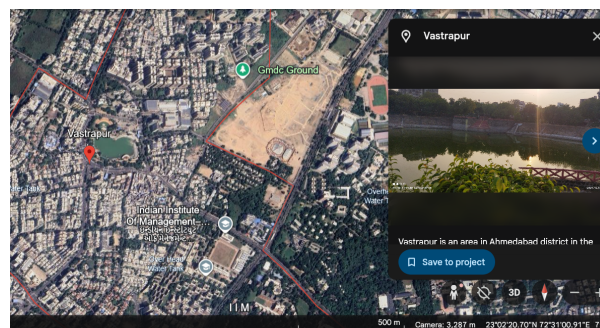


Figure 6: Vastrapur

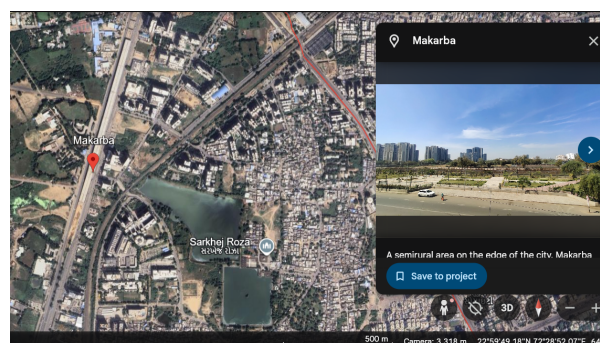


Figure 7: Makabra

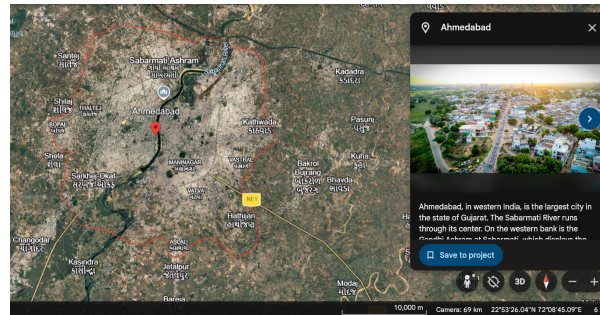


Figure 8: Total Ahmedabad

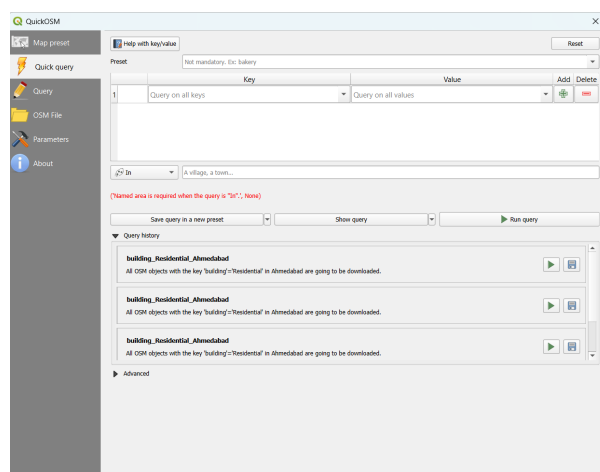


Figure 9: Quick OSM- plugin in qgis

4 Dataset Preparation Workflow

The workflow required for preparing datasets for CityBES simulations was studied in detail.

4.1 Workflow Steps

The workflow understood during this week is:

1. Collect satellite and GIS datasets
2. Import datasets into QGIS
3. Create building footprint polygons
4. Add building attributes
5. Export datasets as GeoJSON
6. Upload datasets into CityBES

4.2 Building Attributes Required

The following building parameters are important for energy simulations:

- Building height
- Number of stories
- Building type
- Year built
- Building geometry

These parameters are necessary for generating accurate urban building energy models.

4.3 GeoJSON Format

GeoJSON format stores:

- Building geometry
- Coordinates
- Building attributes
- Spatial information

GeoJSON acts as the bridge between GIS workflows and CityBES simulations.

5 QGIS Exploration

Initial exploration of QGIS was carried out to understand GIS-based spatial dataset preparation.

5.1 Concepts Explored in QGIS

The following concepts were explored in QGIS:

- GIS layers
- Building footprints
- Spatial datasets
- Coordinate Reference Systems (CRS)
- GeoJSON datasets
- Polygon geometries

5.2 Understanding Developed

QGIS is mainly used for:

- Preparing spatial building datasets
- Managing GIS layers
- Creating building footprint geometries
- Exporting datasets for CityBES

It was understood that QGIS prepares spatial datasets, while CityBES performs building energy simulations.

6 Advanced Concepts Learned

Advanced CityBES dataset preparation concepts were also studied.

6.1 Building Archetypes

Building archetypes represent typical building categories such as:

- Residential buildings
- Office buildings
- Commercial buildings
- Educational buildings

Archetypes help simplify city-scale energy simulations by grouping similar buildings together.

6.2 Advanced Parameters

The following advanced energy parameters were studied:

- Roof insulation R-value
- Window U-value
- SHGC
- HVAC system type
- Cooling COP
- Occupancy schedules

These parameters strongly influence cooling load and building energy performance.

6.3 Schedules in Energy Simulations

Schedules define:

- Occupancy behavior
- HVAC operation timings
- Lighting usage
- Cooling setpoints

Schedules are important because energy consumption changes throughout the day depending on human activity and building usage patterns.

7 Literature Review: Urban Building Energy Modeling for Ahmedabad

As part of the internship work, the research paper titled:

“Assessing a Fit-for-Purpose Urban Building Energy Modelling Framework with Reference to Ahmedabad”

by Anmol Mathur, Pamela Fennell, Rajan Rawal, and Ivan Korolija (2021) was studied.

The paper focuses on Urban Building Energy Modeling (UBEM), a methodology used to estimate and analyze energy consumption at city scale using:

- GIS datasets
- Building characteristics
- Occupancy patterns
- Energy simulation tools

The study highlights the importance of balancing:

- Data availability
- Simulation accuracy
- Computational effort

through a “Fit-for-Purpose” modeling approach.

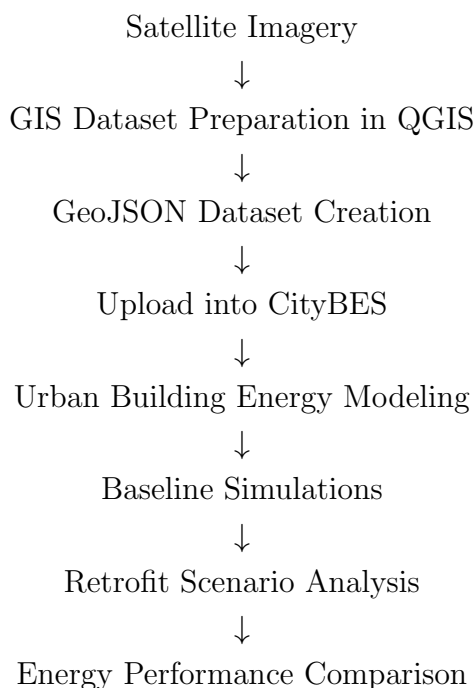
The paper also introduces different Levels of Detail (LoD) for urban energy models, including:

- Building geometry
- Archetypes
- Occupancy
- Climate data
- Urban context

The Ahmedabad case study discussed in the paper is directly related to the internship project and helped in understanding how GIS datasets and building information are transformed into city-scale energy models for retrofit analysis.

8 Overall Understanding

The overall workflow understood during this week involves converting real-world urban buildings into digital building energy models for city-scale retrofit and sustainability analysis.



9 Future Learning Goals

The planned next learning objectives include:

- Learn detailed QGIS workflows
- Understand building footprint extraction
- Study Ahmedabad dataset preparation
- Learn GeoJSON generation workflows
- Explore retrofit scenario implementation in CityBES
- Understand urban energy simulation methods

10 Conclusion

Week 2 mainly focused on understanding the practical workflow of Urban Building Energy Modeling, satellite imagery analysis, GIS dataset preparation, and CityBES-based energy simulations.

The learning developed during this week provided a strong foundation for future work related to:

- Ahmedabad building datasets
- GIS workflows
- Building footprint preparation
- CityBES simulations
- Retrofit analysis

The concepts understood during this week will support future implementation of city-scale retrofit workflows and building energy analysis.